

Data Flow Diagrams (DFD) & System Data Architecture Guide

Context Level 0, Decomposition Level 1, WebRTC Signaling Data Flow, and Ticket Lifecycle State Pipelines

1. Level 0 Context Data Flow Diagram (DFD)

The Level 0 Context Diagram establishes the boundaries of the **NITTE Mentorship & Grievance System (Process 0.0)**. It illustrates the primary external entities (Student, Relationship Officer, Mentor, Admin, YouTube CDN, and Gmail Server) and the high-level data streams flowing across the enterprise boundary.

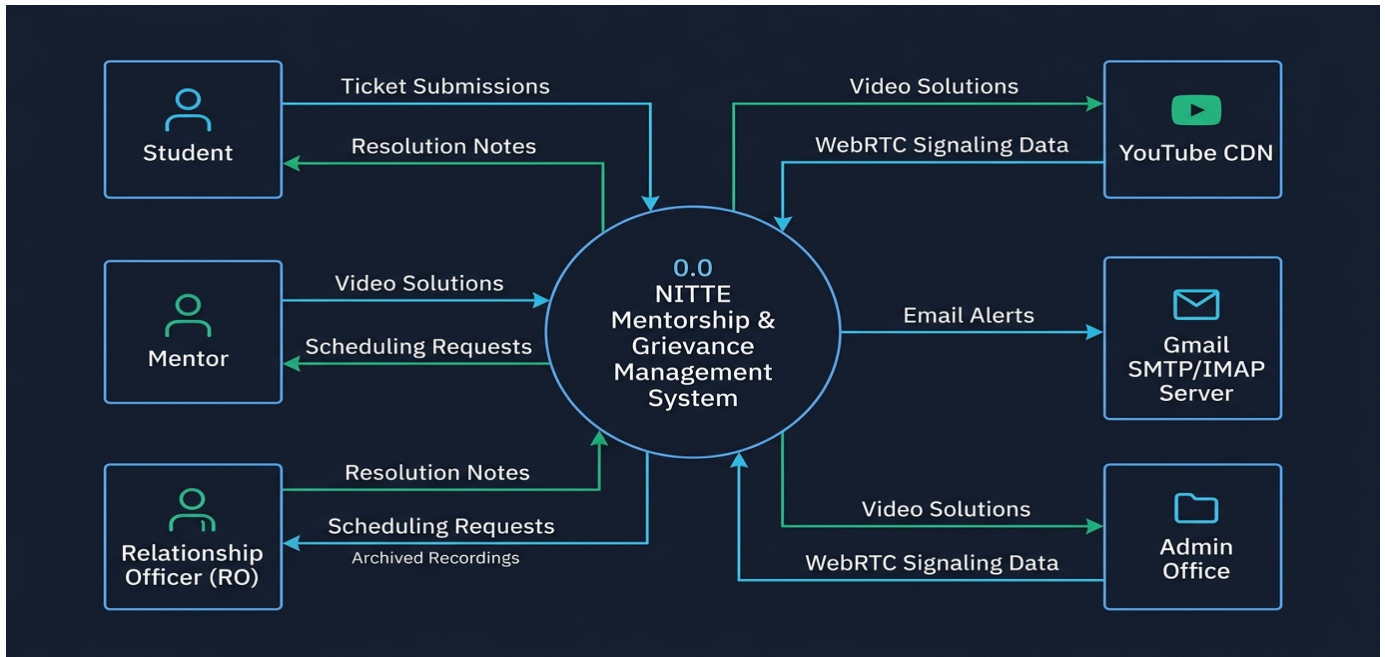


Figure 1: Level 0 Context Data Flow Diagram (External Entities, Inbound Submissions, Signaling Bus & Outbound Media Streams)

External Entity & Boundary Data Stream Dictionary:

External Entity	Inbound Data Stream (To System)	Outbound Data Stream (From System)
Student	Ticket Submissions, Re-open Requests, 1-5 Star Feedback Ratings	Instant Solution Videos, Scheduled Meeting Notices, Resolution Notes
Relationship Officer (RO)	Meeting Schedules, Resolution Actions, Issue-Scoped YouTube Links	Assigned Ticket Queue, Escalation Alerts, Stored Video Recordings
Mentor	Academic Resources, Counseling Notes, Meeting Check-ins	Mentee Activity Feed, Ticket Progress Overviews, Academic Alerts
Admin Office	Administrative Overrides, Role Permissions, System Configs	Auto-Escalated 3rd Attempt Tickets, System Audit Logs, Metrics
YouTube CDN	Embedded Video Frames, Video Metadata (Titles, Thumbnails)	Direct Embed URL Queries via youtube-nocookie.com
Gmail SMTP/IMAP	Inbound Email Replies for Ticket Logs (IMAP Listener)	Instant SMTP Ticket Dispatches, Meeting Invitations & Alerts

2. Level 1 Subsystem Decomposition & Data Store Dictionary

Decomposing the primary system into operational subprocesses reveals how client interactions mutate centralized and client-cached data stores across the PostgreSQL backend and local client state.

Data Store (Repository) Schema Dictionary:

Data Store	Storage Medium	Stored Data Attributes	Associated Processes
D1: users	PostgreSQL / In-Memory	id, name, email, role, region/dept, password	1.0 Auth & Profile Routing
D2: issues	PostgreSQL / LocalStorage	id, studentId, category, priority, status, roId, logs	2.0 Intake, 6.0 Lifecycle
D3: meetings	PostgreSQL / Signals	id, issueId, studentId, roId, date, time, mode, status	3.0 Scheduler, 4.0 Signaling
D4: category_videos	PostgreSQL / LocalStorage	category, title, video_url, description, updated_by	2.0 Intake, 3.0 RO Curation
D5: recordings	PostgreSQL / Blob Store	id, meetingId, issueId, duration, videoUrl (.webm)	5.0 Media Archiver
D6: system_logs	PostgreSQL	id, text, user_role, user_id, timestamp	All Processes (Auditing)

Key Process Data Pipelines:

- **Process 1.0 (Issue Intake & Anti-Spam Gate):** Evaluates 2 tickets / 7-day cooldown rules per student. Valid submissions write to D2: issues and assign to the dedicated RO for that specific category (1 of 49 ROs).
- **Process 2.0 (Multi-Tier Guidance Video Resolver):** Reads D4: category_videos with a 4-tier cascading algorithm: (1) Exact Subcategory Match → (2) Parent Prefix Match → (3) Support Desk Synonym Match → (4) Substring Fallback. Serves instant YouTube solution guidance to student.
- **Process 3.0 (Per-RO Scoped Video Curation):** Enforces that each Relationship Officer can ONLY edit YouTube links for tickets assigned to them in D2. Unauthorized cross-officer updates are denied at the backend with 403 Forbidden.
- **Process 4.0 (Meeting Scheduling & Reassignment Limit):** Manages meeting records in D3. Enforces university rule: maximum 2 meeting reassignments permitted per issue before auto-escalation to Admin.

■ **Real-Time Cross-Tab State Synchronization Pipeline**

To ensure immediate data consistency without high server polling overhead, two dedicated BroadcastChannel event buses operate on the client: (1) 'nitte_video_sync' broadcasts instant updates when an RO publishes a new YouTube guidance link, and (2) 'nitte_meeting_sync' synchronizes call terminations and meeting status changes across open student/RO tabs in **0 milliseconds**.

3. Online Video Meeting & WebRTC Signaling Data Flow (Level 2 DFD)

This Level 2 Data Flow Diagram maps the encrypted real-time multimedia pipeline during an online video call. It traces the handshake exchange through the Signaling Bus down to the browser transceivers, VAD energy analyzers, and automated cloud recording buffers.

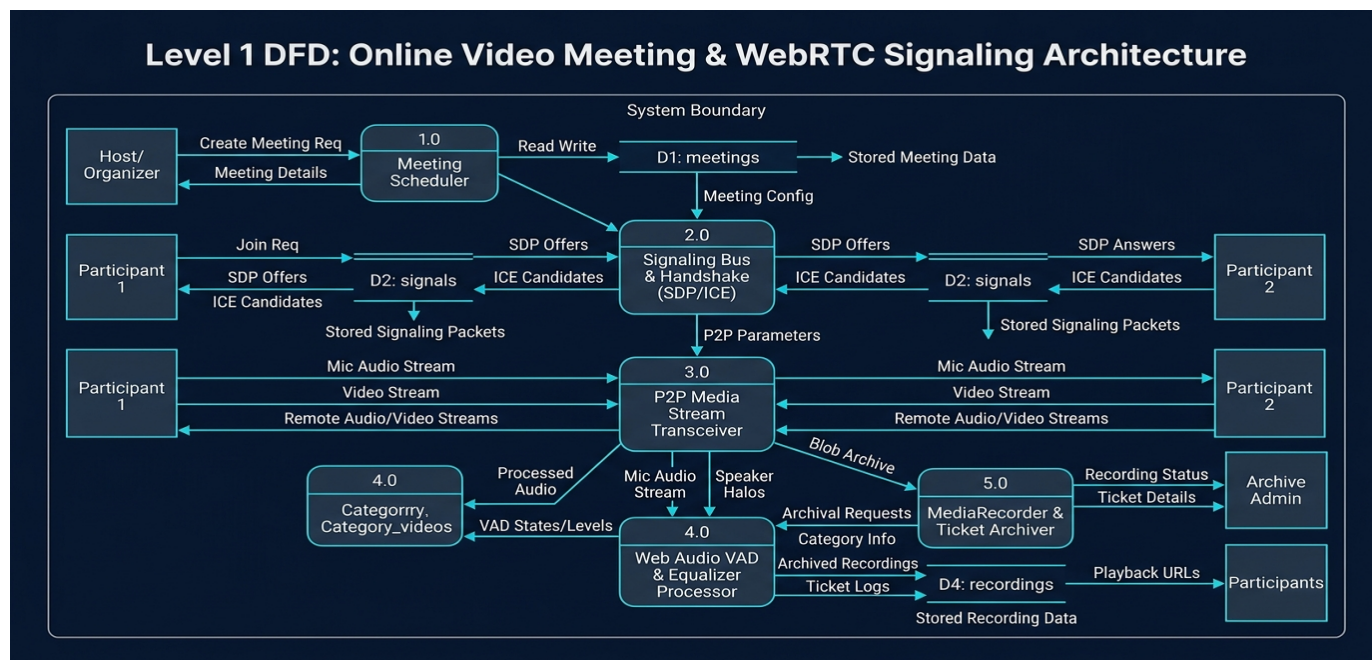


Figure 2: Level 2 DFD — WebRTC P2P Signaling Exchange, Media Stream Transceivers, Web Audio VAD, and Cloud Archiving

WebRTC Signaling Data Packet Schema:

Signal Type	Data Payload Schema	Pipeline Trigger & Purpose
ready	{ type: 'ready', sessionId }	Dispatched upon camera grant; announces peer availability
offer	{ type: 'offer', sdp, sessionId }	RO host creates SDP offer with Opus FEC & 900 kbps clamp
answer	{ type: 'answer', sdp, sessionId }	Student creates SDP answer with explicit receive constraints
candidate	{ type: 'candidate', candidate }	ICE network traversal packets routed via STUN/Direct paths
peer-left	{ type: 'peer-left', sessionId }	Triggers graceful teardown and auto-rejoin readiness
meeting_ended	{ type: 'meeting_ended', issued }	Host signals session conclusion; closes modal on student side

Dual-Pipeline Audio & VAD Data Transformation:

- **Dual Audio Pipeline:** Remote audio is routed directly to a dedicated `<audio autoPlay playsInline />` element to bypass Chromium video rendering pipeline audio locks.
- **Web Audio API RMS Energy Flow:** `MediaStreamTrack` → `AudioContext` → `AnalyserNode` → `Uint8Array TimeDomain Data` → `RMS Calculation` → `React State Hook` → `Emerald Halo UI`. Illuminates when volume exceeds 8%.

4. Ticket Lifecycle State Transition Data Flow

A student grievance moves through a deterministic finite-state pipeline. Every state mutation validates against role permissions, logs an immutable timestamped event into D2: issues.logs, and dispatches real-time email notifications.

Current Status	Triggering Action / Function	Mutated Data Stores	Next Valid States
Submitted	Student clicks 'Submit Issue' (submitIssue())	D2: issues (created) D6: system_logs	Assigned to RO, Resolved (Self-help)
Assigned to RO	Self-help video viewed; student clicks 'Proceed with RO'	D2: issues (status updated) D1: users.ros (routed)	Meeting Scheduled, Escalated
Meeting Scheduled	RO submits date, time, venue in modal (scheduleRoMeeting())	D3: meetings (created) D2: issues (log appended)	Started, Resolved, Escalated
Started	RO launches live video meeting (handleOpenOnlineMeeting())	D3: meetings.status = 'Started' Broadcast signal	Completed
Completed	RO concludes session & saves recording (saveMeetingRecording())	D5: recordings (blob stored) D3: meetings (completed)	Resolved, Escalated
Resolved	RO logs resolution notes (resolveIssue())	D2: issues.status = 'Resolved' Resolution note saved	Re-opened by Student, Closed (Rated)
Re-opened	Student unsatisfied; submits reason (reopenIssue())	D2: issues.status = 'Re-opened' Cooldown preserved	Meeting Scheduled, Escalated (Auto on 3rd)
Escalated	3rd attempt reached OR manual RO escalation (escalateIssue())	D2: issues.status = 'Escalated' Routed to Admin queue	Resolved (Admin override only)
Closed	Student rates experience 1–5 stars and submits feedback	D2: issues.feedback stored	Final State (Archived)

5. Media Archiving & Compliance Data Pipeline

When an online video counseling session concludes, recorded audio/video chunks are serialized into standard .webm blobs on the host client, saved to D5: recordings, and automatically linked to the student's ticket history. Both student and RO can review the recorded counseling video or download the .webm container at any time.